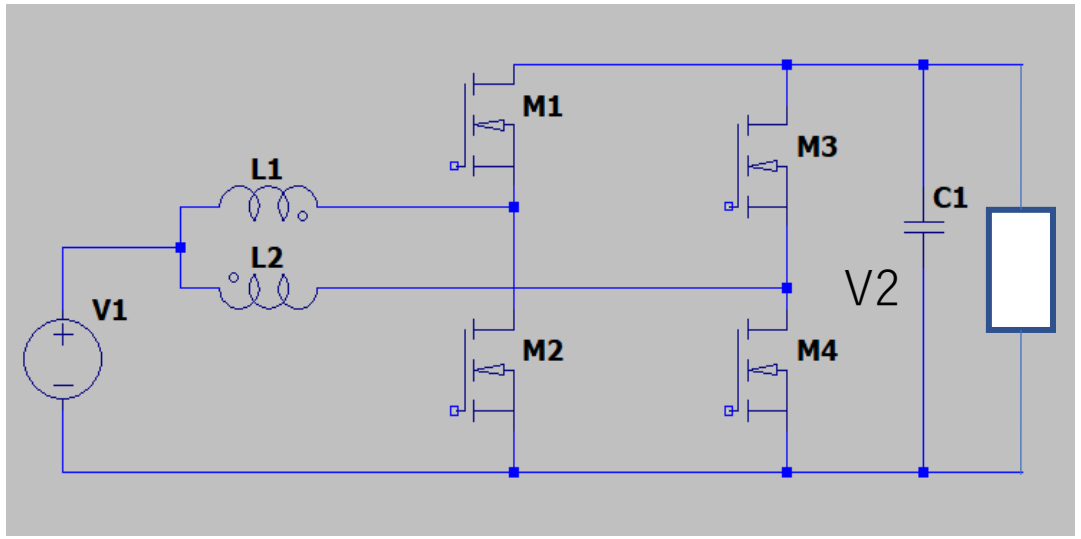
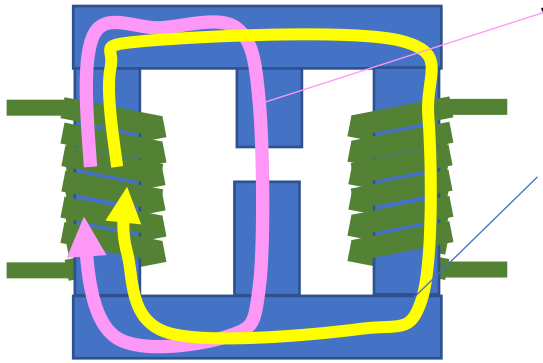
Magnetic coupled DCDC converter





Self inductance(sub magnetic path)

$I1 \times N1 = H1c \cdot \ell 1c + Hg \cdot \ell g$

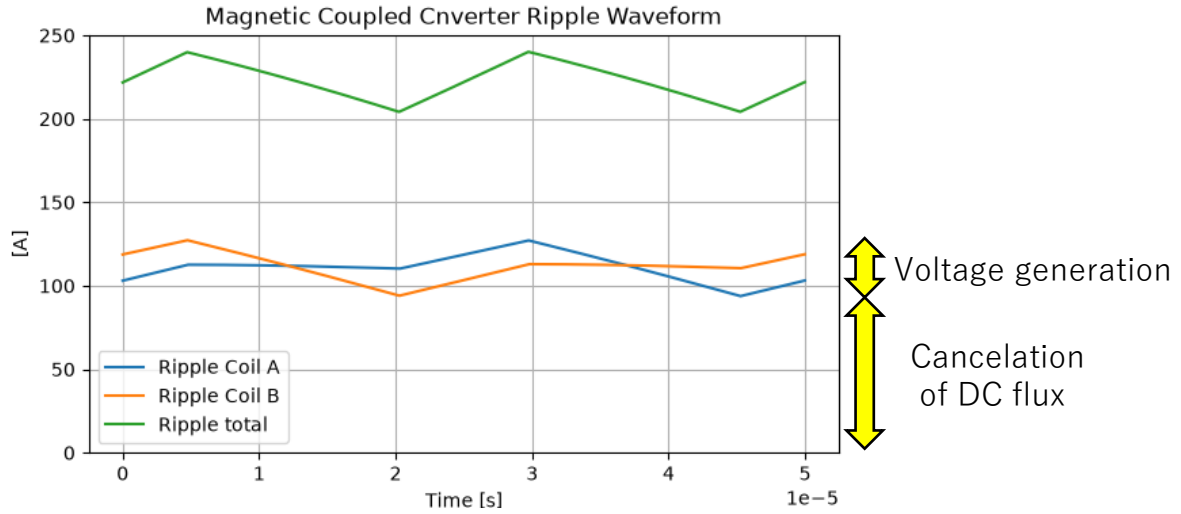
Mutual inductance

$I1 \cdot N1 - I2 \cdot N2 = H2c \cdot \ell 2c$

Simultaneous equations
->can be solved

Cancelation of DC flux with M12
Voltage generation due to self inductance
(High magnetic resistance)
->High powered DCDC converter with small inductor

Mitigate ripple current with interleaved switching



Although linear analysis is not applicable due to the non-linearity of the magnetic material, 1D magnetic circuit simulation via numerical computation is possible because the magnetic path is contained entirely within the core.

Design and Optimization work flow

input

- Operating Conditions
- Geometrical Parameters
- Design Constraints
- Material Properties

DOE Engine(C++)

- Geometry Calculation
- Magnetic Circuit Analysis
- Loss Calculation

output

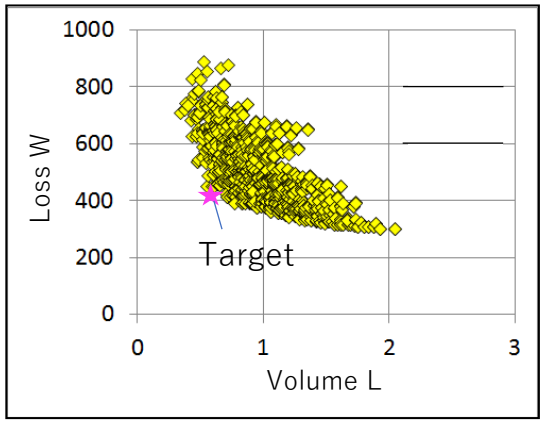
- size
- losses(Iron/Cu)
- L-M-I Characteristics
- Current Waveforms

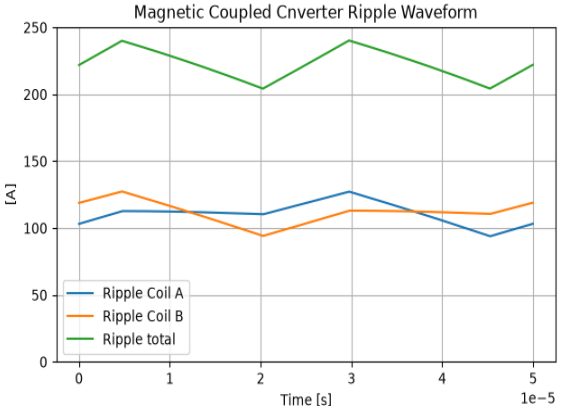
example

	min	max	step	levels
Gap(mm)	0	4	4	2
Turn	18	30	6	3
Iron_h(mm)	25	73	16	4
Iron_w(mm)	25	30	5	2
Cu_w(mm)	4	12	4	3
Cu_thick(mm)	1.3	2.3	1	2
sub-Iron_w(mm)	7	21	7	3
Side core_w(mm)	7	57	10	6
		Total number		5184

1D simulation
DOE Engine

50HR





Nonlinear Magnetic Circuit Analysis Enables Rapid Design Optimization Across Thousands of Design Candidates